

SQLValley / HeriShare - Transactions README

Dataset snapshot: 30 April 2025 | Post-pivot marketplace ledger

1. Purpose of the table

The transactions table records HeriShare's post-pivot whole-asset marketplace activity. Each row represents one attempted or completed transaction for a cultural asset. The table begins with the quiet marketplace reopening in late September 2024 and intentionally contains no pre-pivot fractional-share transactions.

Its main role is to connect the commercial history of the platform across three other core tables: accounts, products, and employees. Completed transactions also form the ownership history used to determine the current owner of a product.

2. Canonical schema

Field	Type / FK	Meaning
t_id	INTEGER PRIMARY KEY	Unique transaction identifier.
vendor	TEXT -> accounts(username)	Account selling the product.
buyer	TEXT -> accounts(username)	Account attempting to buy the product.
prod_id	INTEGER -> products(p_id)	Product involved in the transaction.
date_time	TEXT	Timestamp of the transaction record.
price	REAL	Agreed or attempted whole-asset price.
validated_by	INTEGER -> employees(e_id)	Certified employee who manually processed the transaction; may be NULL when no manual processing occurred.
status	TEXT	Operational outcome of the transaction.

The SQL export contains only these canonical fields. Navigation/helper fields are present only in the CSV/XLSX generation outputs.

3. Dataset scale

Metric	Value
Total transaction rows	1,800
Completed transactions	1,500
Distinct products with completed sales	820
Completed GMV	\$1,132,362,500
Date range	29 Sep 2024 - 30 Apr 2025

Completed-sale frequency by product

Completed sales per product	Products
1	300
2	380
3	120
4	20

Most traded assets change hands only once or twice. A smaller liquid subset generates repeat marketplace activity. This creates meaningful ownership chains without turning the platform into a high-frequency trading system.

4. Transaction statuses

Status	Rows	Meaning	Ownership effect
COMPLETED	1,500	Certified employee processed the sale successfully.	Buyer becomes owner.
PENDING	70	Transaction exists but manual processing is not yet complete.	None yet.

Status	Rows	Meaning	Ownership effect
CANCELLED	80	Buyer or vendor withdrew before completion.	None.
REJECTED	65	Certified employee reviewed and declined the transaction.	None.
FAILED	55	Transaction could not settle or complete operationally.	None.
REVERSED	30	Previously processed transaction was undone.	No lasting transfer in the final ownership state.

Only COMPLETED rows transfer ownership. Failed, cancelled, rejected, pending, and reversed rows remain in the ledger for operational history and SQL exercises.

5. Meaning of validated_by

validated_by does not mean a transaction was merely assigned to an employee. It identifies the certified HeriShare employee who manually processed the transaction.

- Every COMPLETED transaction has a valid certified employee.
- REJECTED transactions may also have a validator because rejection can be the result of manual review.
- PENDING or early-cancelled attempts may have validated_by = NULL.
- The referenced employee must already be employed on the transaction date.

6. Helper fields in CSV/XLSX

Helper	Meaning
phase	Story/business phase in which the transaction occurs. All generated transactions are Phase 4.
event	Nearest mainline story event controlling the transaction environment.
month	YYYY-MM grouping for easy filtering and monthly aggregation.
product_tx_no	Sequence number of all transaction attempts for that product.
completed_transfer_no	Sequence number among completed ownership transfers for that product; blank when not completed.
is_resale	TRUE when the vendor acquired the product through an earlier completed marketplace transaction.
is_relevant	TRUE for storyline-relevant transactions/products.

These helpers exist only to make the dataset easier to inspect and teach from. They are not part of the canonical transactions schema.

7. Timeline and traffic pattern

Period	Rows	Completed	Business effect
29-30 Sep 2024	16	8	Controlled reopening; high manual-review friction.
Oct 2024	115	92	Early marketplace trading; original suppliers dominate.
Nov 2024	180	145	Confidence improves; first meaningful resales.
Dec 2024	225	185	Year-end collection and auction activity increases volume.
Jan 2025	275	230	Repeat buyers and seller/buyer role overlap increase.
Feb 2025	298	250	Processing standardizes; certified validator pool expands.
Mar 2025	335	285	Secondary-market liquidity becomes significant.
Apr 2025	356	305	Highest-volume month in the snapshot.

Most rows represent ordinary marketplace traffic rather than named story beats. Events influence transaction volume, rejection rates, resale frequency, and operating conditions; they do not dictate every individual sale.

8. Foreign-key and ownership logic

- vendor and buyer always reference existing accounts(username) values.
- vendor and buyer are never the same account in a single transaction.
- prod_id always references an existing products(p_id).
- Non-NULL validated_by values always reference employees(e_id).
- For a completed sale, vendor must be the product owner immediately before that sale.

- The buyer becomes the owner only after a COMPLETED transaction.
- For a resale, the previous completed buyer becomes the next completed vendor.
- After replaying all completed transfers, products.owned_by equals the latest valid owner.

Products reconciliation

Because the products dataset was generated before the detailed transaction ledger, a reconciled products export is provided. It updates owned_by and current product status after replaying the transaction history. The transactions table stores history; the products table stores current state.

9. Story-relevant transactions

The five Plot Hole products have explicit post-pivot transaction histories and are marked through is_relevant in the navigation output. Their rows are still shaped like ordinary marketplace records: they use valid accounts, products, prices, and certified employees rather than special story-only fields.

This keeps the narrative discoverable through joins and filters instead of making the answer obvious from the raw schema.

10. Validation guarantees

- Exactly 1,800 transaction rows and 1,500 COMPLETED rows.
- All primary keys are unique.
- All account, product, and non-NULL employee foreign keys resolve.
- All completed transactions have certified manual validation.
- Transaction participants exist before the relevant transaction timestamp.
- No transaction predates the post-pivot reopening.
- Completed ownership chains are chronologically consistent.
- Unsuccessful transaction statuses do not create ownership transfers.
- Final product ownership matches the latest valid completed transfer.
- Prices are positive and generated relative to product value and prior sale history.

11. Generated files

File	Purpose
sqlvalley_transactions.csv	Full teaching/navigation dataset including helper fields.
sqlvalley_transactions.xlsx	Transaction data plus monthly, status, validator, product-trade, important-transaction, and validation sheets.
sqlvalley_transactions.sql	Canonical CREATE TABLE and INSERT statements without helper fields.
sqlvalley_products_reconciled.csv	Products after ownership/status reconciliation.
sqlvalley_products_transaction_reconciliation.sql	SQL updates required to reconcile products with the generated ledger.

12. Intended SQL exercise use

The table is designed to support beginner-to-intermediate exercises involving:

- Filtering by status, date, price, or validator.
- COUNT, SUM, AVG, MIN, MAX, GROUP BY, and HAVING.
- Monthly transaction and GMV trends.
- Joining transactions to accounts, products, and employees.
- Finding top buyers, vendors, validators, or product categories.
- Identifying repeat buyers and resold products.
- Tracing ownership chains for a product.
- Finding products that never completed a sale.

- Comparing successful and unsuccessful transaction attempts.
- Validating whether products.owned_by agrees with transaction history.